PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Technologies selected for the existing F1 analytics dashboard and website

Date	31 January 2026	**Team ID**	
Project Name	F1 Racing Analytics	**Maximum Marks**	2 Marks

The project combines structured F1 dataset files, Tableau for business intelligence and a Bootstrap-based website for presenting the dashboard and story. The entries below reflect the project as described; tools not evidenced in the project are not added as implemented technologies.

S.No	Component	Description	Technology	Justification / Purpose
1	Programming / Query Language	Used to build the website and work with structured F1 data.	HTML, CSS, JavaScript	HTML structures the pages, CSS styles the responsive interface, and JavaScript supports navigation and Tableau embed loading.
2	Database / Data Warehouse	Stores and manages the structured data used for analysis.	CSV / text dataset files	The project data is supplied as files and connected to Tableau; a separate database or warehouse is not part of the confirmed implementation.
3	Data Ingestion	Loads source data into the analytical environment.	Tableau data connections; multiple text files	Allows the F1 dataset files to be connected and analysed in Tableau. Files with matching columns can be combined using a union where appropriate.
4	Data Transformation and Business Logic	Cleans, joins, filters, aggregates and calculates measures.	Tableau Desktop / Tableau calculations	Supports points aggregation, driver and team comparisons, KPI calculations, and Top N filtering for the driver leaderboard story.
5	Data Visualization / Analytics	Creates dashboards, charts, KPIs and visual analysis.	Tableau Public	Presents driver leaderboard, team performance, championship analysis, tyre compounds, country participation, podiums and DNF insights.
6	Front-End / User Interface	Interface through which users access dashboards and story.	HTML, CSS, Bootstrap, JavaScript; Tableau Public embeds	Provides Home, Overview, Dashboard and Story navigation while embedding the published Tableau visualisations.

Architecture summary

F1 source files (CSV/text) → Tableau data connection and calculations → Tableau Public dashboard/story → Bootstrap website pages that embed the published views.

Scope note: No SQL database, Python processing pipeline, Flask or Streamlit app is assumed in this version because those technologies were not confirmed as part of the supplied project implementation.